# Activity 6.2: Exponential Functions, Growth & Decay

**A**

Name: _____ Partner's name: _____ Date: _____

**Directions:** Evaluate, identify, or write each exponential function as directed. Show your work.

*Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.*

**1** If $f(x) = 2^x$, find $f(5)$.

**Answer:** _____ Match? ☐ Yes ☐ No

**2** If $f(x) = 3(2)^x$, find $f(4)$.

**Answer:** _____ Match? ☐ Yes ☐ No

**3** Is $y = 200(1.06)^x$ growth or decay? What is the percent rate?

**Answer:** _____ Match? ☐ Yes ☐ No

**4** Is $y = 100(0.85)^x$ growth or decay? What is the percent rate?

**Answer:** _____ Match? ☐ Yes ☐ No

**5** A population of 500 grows 10% each year. Write an equation for the population $y$ after $t$ years.

**Answer:** _____ Match? ☐ Yes ☐ No

**6** A car bought for \$5,000 loses 20% of its value each year. What is it worth after 2 years?

**Answer:** _____ Match? ☐ Yes ☐ No

**7** A bacteria culture starts at 100 and doubles every hour. How many bacteria are there after 4 hours?

**Answer:** _____ Match? ☐ Yes ☐ No

**8** Evaluate $f(x) = 4(3)^x$ at $x = 2$.

**Answer:** _____ Match? ☐ Yes ☐ No

# Activity 6.2: Exponential Functions, Growth & Decay

**B**

Name: _____ Partner's name: _____ Date: _____

**Directions:** Evaluate, identify, or write each exponential function as directed. Show your work.

*Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.*

**1** If $f(x) = 4(2)^x$, find $f(3)$.

**Answer:** _____ Match? ☐ Yes ☐ No

**2** If $f(x) = 6(2)^x$, find $f(3)$.

**Answer:** _____ Match? ☐ Yes ☐ No

**3** Is $y = 50(1.06)^x$ growth or decay? What is the percent rate?

**Answer:** _____ Match? ☐ Yes ☐ No

**4** Is $y = 40(0.85)^x$ growth or decay? What is the percent rate?

**Answer:** _____ Match? ☐ Yes ☐ No

**5** An investment of \$500 grows 10% each year. Write an equation for its value $y$ after $t$ years.

**Answer:** _____ Match? ☐ Yes ☐ No

**6** A radioactive sample has a mass of 12,800 mg and loses half its mass every hour. What is its mass after 2 hours?

**Answer:** _____ Match? ☐ Yes ☐ No

**7** A bacteria culture starts at 25 and doubles every hour. How many bacteria are there after 6 hours?

**Answer:** _____ Match? ☐ Yes ☐ No

**8** An exponential function has $f(0) = 4$ and $f(1) = 12$. Find $f(2)$.

**Answer:** _____ Match? ☐ Yes ☐ No